Logical Order of the Proof

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The goal is to prove, for all multi-indices β that :

$$\begin{cases} \mathbb{E}^{\frac{1}{p}} \left|\Pi_{x\beta r}(x)\right|^p \lesssim r^{|\beta|}, \\ \mathbb{E}^{\frac{1}{p}} \left|\Pi^-_{x\beta r}(x)\right|^p \lesssim r^{|\beta|-2}, \\ \mathbb{E}^{\frac{1}{p}} \left|(\Gamma^*_x)^\gamma_\beta\right|^p \lesssim x^{|\beta|-|\gamma|} \quad \text{for all populated } \gamma, \end{cases}$$

where the implicit constants only depend on d, s, β, γ, p, on ψ only through the semi-norms, but not on x, r, and ρ.

To do, we proceed by induction on the multi-indices β, with the order $|\cdot|_\prec$. Furthermore, we need to enrich our induction hypothesis. For β a multi-index, let our set of hypothesis be :

- Target bounds :

$$\begin{cases} \mathbb{E}^{\frac{1}{p}} \left|\Pi_{x\beta r}(x)\right|^p \lesssim r^{|\beta|}, & (1) \\ \mathbb{E}^{\frac{1}{p}} \left|\Pi^-_{x\beta r}(x)\right|^p \lesssim r^{|\beta|-2}, & (2) \\ \mathbb{E}^{\frac{1}{p}} \left|(\Gamma^*_x)^\gamma_\beta\right|^p \lesssim x^{|\beta|-|\gamma|} \quad \text{for all populated } \gamma, & (3) \end{cases}$$

- Coherence bound : (the denomination will be clearer later)

$$\begin{aligned} &\text{for all populated } \gamma, \quad \left(\int_{B_R} dx\, \mathbb{E}^{\frac{2}{q}} \left|\left(\mathrm{d}\Gamma^*_{x+y} - \mathrm{d}\Gamma^*_x \Gamma^*_{x\,x+y}\right)^\gamma_\beta\right|^q\right)^{\frac{1}{2}} \\ &\lesssim \begin{cases} |y|^{\alpha-|\gamma|_p+\frac{D}{2}} (|y|+R)^{|\beta|-|\gamma|+|\gamma|_p-\alpha}, & \text{if } \alpha - |\gamma|_p + \frac{D}{2} > 0, \\ (|y|+R)^{|\beta|-|\gamma|+\frac{D}{2}}, & \text{else.} \end{cases} \end{aligned} \tag{4}$$

- Estimate on coefficients:

$$\text{for all populated } \gamma, \quad \left(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} \left|(\mathrm{d}\Gamma^*_x)^\gamma_\beta\right|^q\right)^{\frac{1}{2}} \lesssim R^{\frac{D}{2}+|\beta|-|\gamma|}, \tag{5}$$

The proof relies on the spectral gap inequality :

$$\mathbb{E}^{\frac{1}{p}} \left|\Pi^-_{\beta r}(0)\right|^p \lesssim \left|\mathbb{E}\Pi^-_{\beta r}(0)\right| + \sup\{\mathbb{E}^{\frac{1}{q}} \left|\delta\Pi^-_{\beta r}(0)\right|^q ; \delta\xi \text{ random with } \mathbb{E}\left\|\delta\xi\right\|^{p^*}_{\dot{H}_s} \leq 1\}, \tag{6}$$

for $1 \leq q < p^* \leq 2 \leq p$. We will prove estimates for all p, q, p^* in this range so we will be able to use the induction hypothesis with $p' \geq p$ and $p'^* > q' \geq q$ as it naturally occurs with Hölder's inequality.

We claim that we can control $\left|\mathbb{E}\Pi^-_{\beta r}(0)\right|$ when $|\beta| < 2$. We need to control the second term.

Sketch of the reasoning :

A promising strategy is to express the increments as follows:

$$\begin{aligned}\Pi^-_{x+y\beta} - \Pi^-_{x\beta} &= ((\mathrm{Id} - \Gamma^*_{xx+y})\Pi^-_{x+y})_\beta \\ &= \sum_\gamma (\mathrm{Id} - \Gamma^*_{xx+y})^\gamma_\beta \Pi^-_{x+y\gamma} \\ &= -\sum_{\gamma \prec \beta} (\Gamma^*_{xx+y})^\gamma_\beta \Pi^-_{x+y\gamma} \\ &= -\sum_{\substack{\gamma \prec \beta \\ \gamma \neq \mathrm{pp}}} (\Gamma^*_{xx+y})^\gamma_\beta \Pi^-_{x+y\gamma},\end{aligned}$$

taking advantage of the triangularity of the operator $\mathrm{Id} - \Gamma^*_{xx+y}$ with respect to $\prec$, and the fact that $\gamma = \mathrm{pp} \implies \Pi^-_{z\gamma} = 0$. As a result, we can use the induction hypothesis (2) to estimate $\Pi^-_{x+y\gamma}$.

In addition, for all $z \in \mathbb{R}^d$ and $\gamma \neq \mathrm{pp}$,

$$(\Gamma^*_z)^\gamma_\beta = \sum_{k\delta_3 + \beta_1 + \cdots + \beta_l = \beta} (\Gamma^*_z)^{\delta_{\mathbf{n}_1}}_{\beta_1} \cdots (\Gamma^*_z)^{\delta_{\mathbf{n}_l}}_{\beta_l}, \tag{7}$$

and $k\delta_3 + \beta_1 + \cdots + \beta_l = \beta \implies \forall i \in \{1, \dots, l\}$, $\beta_i \prec \beta$, so that (3) is true for $\gamma \neq \mathrm{pp}$:

$$\begin{aligned}\mathbb{E}^{\frac{1}{p}} \left|(\Gamma^*_z)^\gamma_\beta\right|^p &\lesssim \sum \mathbb{E}^{\frac{1}{lp}} \left|(\Gamma^*_z)^{\delta_{\mathbf{n}_1}}_{\beta_1}\right|^{pl} \cdots \mathbb{E}^{\frac{1}{lp}} \left|(\Gamma^*_z)^{\delta_{\mathbf{n}_l}}_{\beta_l}\right|^{pl} \\ &\lesssim z^{|\beta_1| + \cdots + |\beta_1| - |\mathbf{n}_1| - \cdots - |\mathbf{n}_l|} \\ &\lesssim z^{|\beta| - |\gamma|}.\end{aligned}$$

Fix $\beta \neq 0$ such that $\forall \gamma \prec \beta$, (P_γ) holds. Fix p, q such that $1 \leq q < p^* \leq 2 \leq p$.

From (4), we deduce that the germ $(\delta\Pi^-_\beta - (\mathrm{d}\Gamma^*_x \Pi^-_x)_\beta)$ is coherent. By reconstruction, we get :

$$\left(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} \left|(\delta\Pi^- - \mathrm{d}\Gamma^*_x \Pi^-_x)_{\beta r}(x)\right|^q\right)^{\frac{1}{2}} \lesssim r^{\alpha - 2 + \frac{D}{2}} (r + R)^{|\beta| - |\alpha|}. \tag{8}$$

By (5) and (2), the germ $(\mathrm{d}\Gamma^*_x \Pi^-_x)_\beta$ is homogeneous, so we get by triangle inequality an estimate on $\delta\Pi^-$:

$$\left(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} \left|\delta\Pi^-_{\beta r}(x)\right|^q\right)^{\frac{1}{2}} \lesssim r^{\alpha - 2 + \frac{D}{2}} (r + R)^{|\beta| - |\alpha|}.$$

By Sobolev's inequality, we can post-process this inequality to obtain the pointwise estimate :

$$\mathbb{E}^{\frac{1}{q}} \left|\delta\Pi^-_{\beta r}(x)\right|^q \lesssim r^{\alpha - 2 - \frac{D}{2}} (r + |x|)^{|\beta| - \alpha + \frac{D}{2}}. \tag{9}$$

We have the spectral gap inequality :

$$\mathbb{E}^{\frac{1}{p}} \left|\Pi^-_{\beta r}(0)\right|^p \lesssim \left|\mathbb{E}\Pi^-_{\beta r}(0)\right| + \sup\{\mathbb{E}^{\frac{1}{q}} \left|\delta\Pi^-_{\beta r}(0)\right|^q ; \delta\xi \text{ random with } \mathbb{E}\left\|\delta\xi\right\|^{p^*}_{\dot{H}_s} \leq 1\}, \tag{10}$$

and for $|\beta| < 2$, we have the control :

$$\left|\mathbb{E}\Pi^-_{\beta r}(0)\right| \lesssim r^{|\beta| - 2}.$$

Estimate (9) at $x = 0$ with these last two estimates gives :

$$\mathbb{E}^{\frac{1}{p}} \left| \Pi^-_{\beta r}(0) \right|^p \lesssim r^{|\beta|-2}, \tag{11}$$

which generalizes to the target estimate (2) because $\Pi^-_x(\cdot + x) \overset{\text{law}}{=} \Pi^-$. For $|\beta| > 2$, we need estimate (3) for $\gamma \neq$pp, which we prove by an algebraic argument

By integration, we deduce target estimate (1)